

*Figure 5-3: The divide and merge phases of merge sort*

For example, at the end of the division phase, we have eight separate lists of single numbers: [2], [9], [8], [5], [3], [4], [7], [6]. A list of just one number is naturally in sorted order. Merging two sorted lists into a larger sorted list involves looking at the start of both smaller lists and appending the smaller value to the larger list. [Figure 5-4](#) shows an example of merging [2, 9] and [5, 8]. This is repeatedly done in the merge phase until the end result is that the original `mergeSort()` call returns the full list in sorted order.